Thesis for Doctor of Philosophy

Preparation and Mechanical Properties of TPS(Thermoplastic Starch)/Poly(Butylene Succinate) Blends

June 2026

Graduate School
Kumoh National Institute of Technology

Department of Mechanical System Engineering

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Supervisor Kim Gil Dong

This Thesis is Presented for the Doctor of Philosophy

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Preparation and Mechanical Properties of TPS(Thermoplastic Starch)/Poly(Butylene Succinate) Blends

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Abstract

Polymer materials are widely used in industry nationwide; however, most of these are non-biodegradable, and it is very difficult to manage their waste after use. These wastes become white pollution as a worldwide environmental problem. Development of a biodegradable polymer is one of the solutions to abate the aforementioned problem. Biodegradable polymers can be developed by combining biodegradable matrices and biodegradable fillers.

Keywords: thermoplastic starch (TPS); poly(butylene succinate) (PBS); biodegradable blends

TPS(Thermoplastic Starch)와 Poly(Butylene Succinate)의 Blends 제조와 물리적 특성

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국립금오공과대학교

요 약

본 연구에서는 대표적인 천연 고분자의 하나인 전분을 여러 가소제를 사용 하여 가소화 시킨 thermoplastis starch(TPS)를 제조하여 이를 지방족 폴리에 스터의 하나인 polybutylene succinte(PBS)와 혼합비를 달리하며 블렌드 하 였다. TPS의 조성 및 함량이 PBS의 열적 특성 및 물리적 특성에 미치는 영향 에 대해 확인하였다. 본 연구에서는 대표적인 천연 고분자의 하나인 전분을 여러 가소제를 사용 하여 가소화 시킨 thermoplastis starch(TPS)를 제조하여 이를 지방족 폴리에 스터의 하나인 polybutylene succinte(PBS)와 혼합비를 달리하며 블렌드 하 였다. TPS의 조성 및 함량이 PBS의 열적 특성 및 물리적 특성에 미치는 영향 에 대해 확인하였다. fillers.

Keywords: thermoplastic starch (TPS); poly(butylene succinate) (PBS); biodegradable blends

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REMARK

Material from:

Some passages of this dissertation have been reprinted from the above sources.

Chapter 1

INTRODUCTION AND BACKGROUND

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1.1 Section

This is a section in Chapter 1.

1.1.1 Example Subsection

This is a subsection in Chapter 1.

Example Subsubsection

This is a subsubsection in Chapter 1.

1.2 Citation

Citations are **ref1**, **ref2**, **ref3**, **ref4**, **einstein**, **knuth-fa** and **dirac**.

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CHAPTER 2

METHODS

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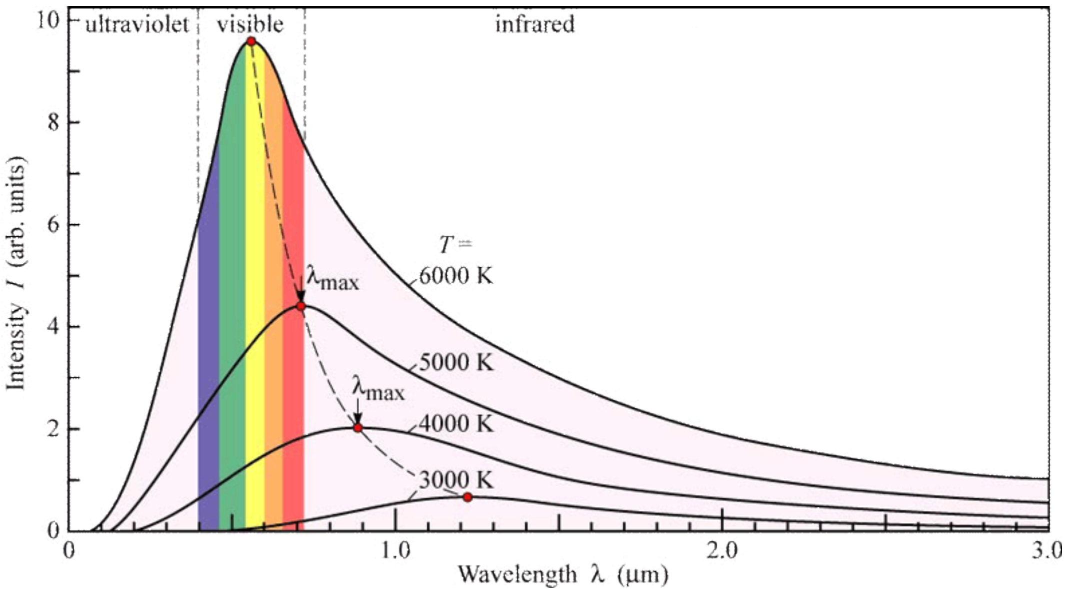

Figure 2.1: Portrait figure

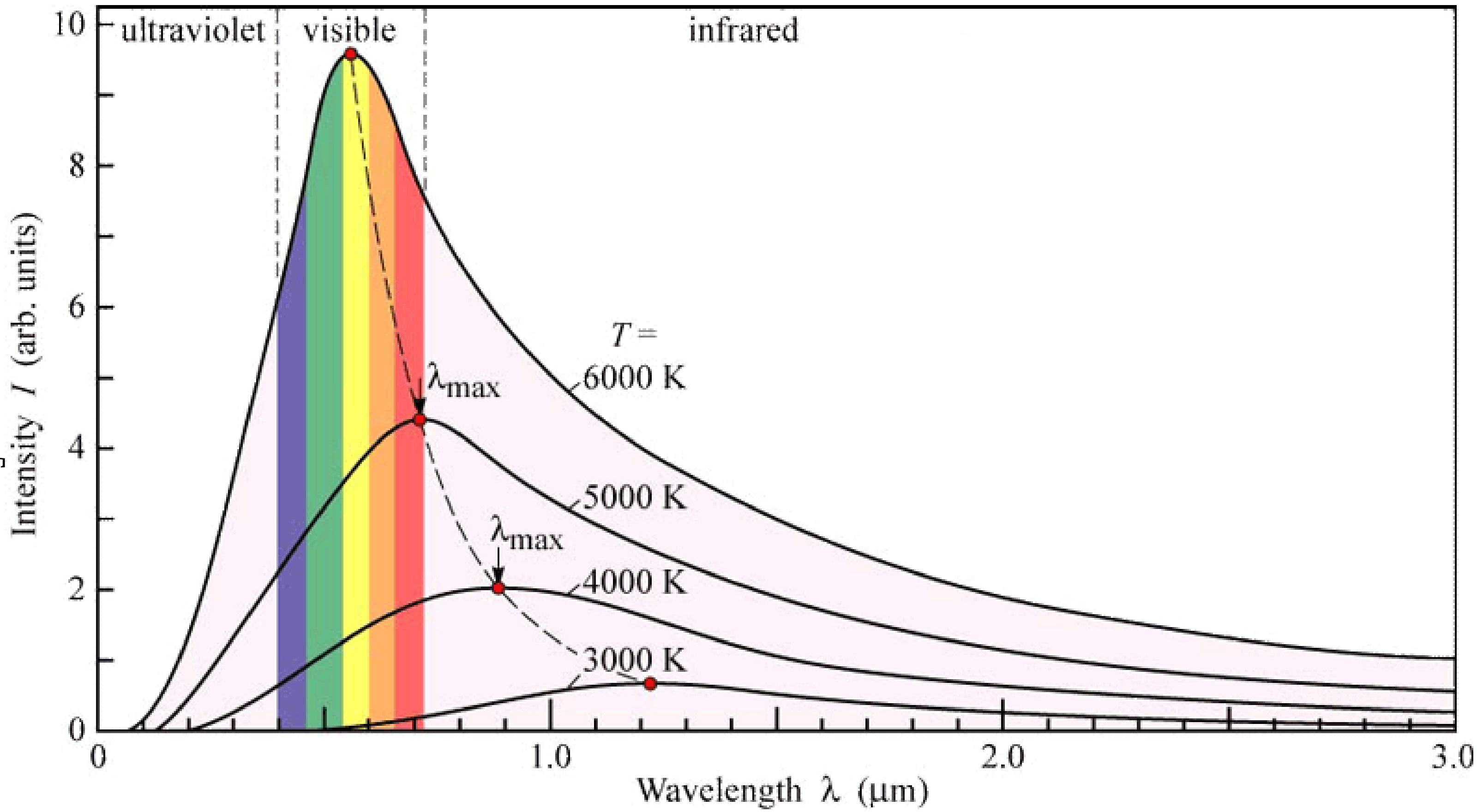

Figure 2.2: Landscape figure

CHAPTER 3

RESULTS

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Table can be cited as Table 3.1 or Table 3.2. The portrait table is Table 3.1, and landscape table is Table 3.2.

x	f(x)	g(x)
1	6	4
2	6	3
3	6	2
4	6	2

Table 3.1: This is an example portrait table.

x	f(x)	g(x)	A	B	C	D	E	F	G	H	I	J
2	6	3	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
1	6	4	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
3	6	2	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
4	6	2	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0

Table 3.2: This is an example landscape table.

CHAPTER 4

DISCUSSION

Equation is cited as Equation (4.1) and Equation (4.3) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat **ref1**. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur **ref2**. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum **ref3**.

$$F = ma \tag{4.1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f} \tag{4.2}$$

$$\nabla \cdot \mathbf{u} = 0 \tag{4.3}$$

CHAPTER 5

CONCLUSION

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Appendices

Appendix A

EXPERIMENTAL EQUIPMENT

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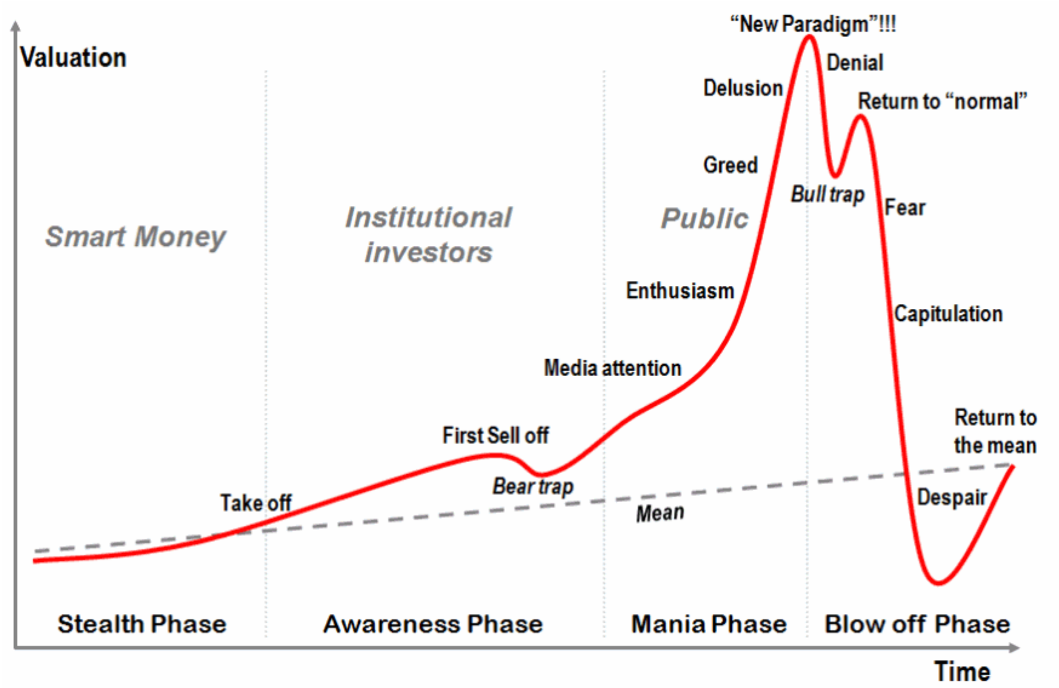

Figure A.1: Stage of a bubble

APPENDIX B

DATA PROCESSING

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Monospace font Test

`PPO Algorithm in Pseudocode.`

Bold and Monospace Test

`PPO Algorithm in Pseudocode.`

Algorithm 1 PPO

for $iteration = 1, 2, \ldots$ **do**
  for $actor = 1, 2, \ldots, N$ **do**
    Run policy $\pi_{\theta_{old}}$ in environment for T time steps
    Compute advantage estimates $\hat{A}_1, \ldots, \hat{A}_T$
  end for
  Optimize surrogate L wrt. θ, with K epochs and minibatch size $M \leq NT$
  $\theta_{old} \leftarrow \theta$
end for